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Occurrence of per- and polyfluoroalkyl substances (PFAS) in commercially available bottled drinking water in Saudi Arabia

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ABSTRACT

Background: Per- and polyfluoroalkyl substances (PFAS) are persistent environmental contaminants of increasing concern due to their widespread occurrence, environmental persistence, and potential adverse health effects. Monitoring PFAS in drinking water is important for consumer safety and quality assurance, particularly because water is consumed directly and is widely used as an ingredient or processing medium in foods and beverages.

Objective: This study aimed to investigate the occurrence and concentrations of selected PFAS in commercially available bottled drinking water in Saudi Arabia and to provide preliminary regional data relevant to contaminant surveillance and exposure assessment.

Methods: Nineteen commercially available bottled drinking-water brands were purchased from retail markets in Riyadh,

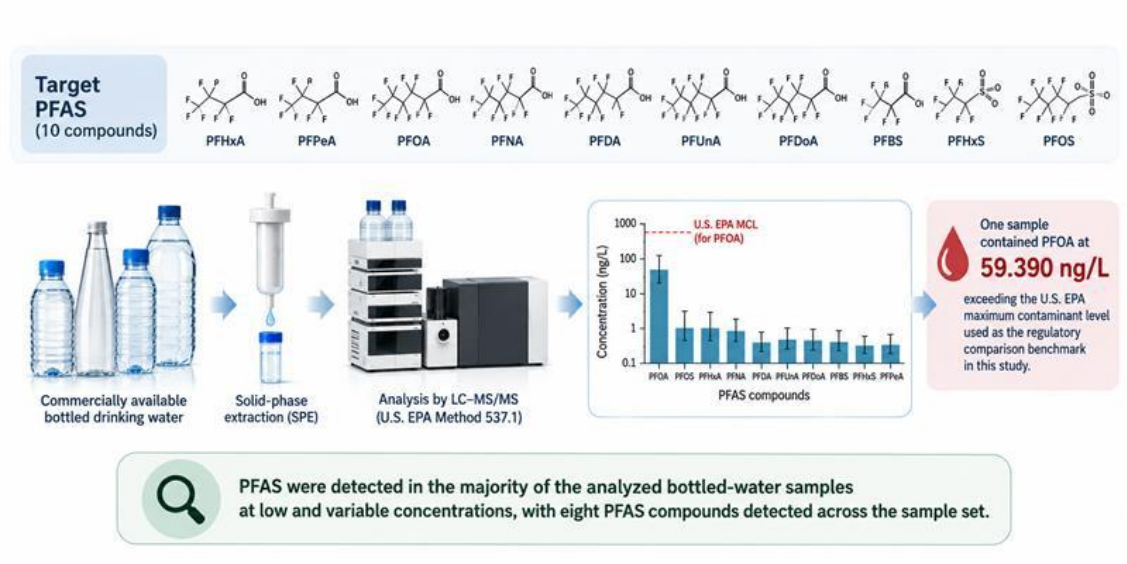
Saudi Arabia, and analyzed for selected PFAS using solid-phase extraction coupled with liquid chromatography-tandem mass spectrometry (SPE-LC-MS/MS), based on U.S. EPA Method 537.1. Analytical performance was evaluated through calibration linearity, repeatability, recovery, limits of detection and quantification, and quality-control samples. The analytical procedure provided adequate performance for several target PFAS, although variable recoveries were observed for some analytes and were considered when interpreting the reported concentrations.

Results: PFAS were detected in the majority of the analyzed bottled-water samples at low and variable concentrations, with eight PFAS compounds detected across the sample set. One sample contained perfluorooctanoic acid (PFOA) at 59.390 ng/L, exceeding the U.S. EPA maximum contaminant level used as the regulatory comparison benchmark in this study.

Conclusions: The occurrence of PFAS in the majority of the analyzed bottled-water samples highlights the importance of continued surveillance of persistent contaminants in drinking water. The findings provide preliminary regional evidence that may support future PFAS monitoring, exposure and risk-assessment efforts, and water-quality assurance initiatives relevant to consumer and food safety.

Novelty: The principal contribution of this study is the generation of regional occurrence data on PFAS in commercially available bottled drinking water in Saudi Arabia, rather than the development of a new analytical method. These data help address the limited information available on PFAS occurrence in bottled drinking water in the region and provide a basis for future monitoring and exposure-assessment efforts.

Keywords: PFAS; bottled drinking water; Saudi Arabia; PFOA; LC-MS/MS; contaminant monitoring; water quality; consumer safety



Graphical Abstract. PFAS Occurrence in Commercially Available Bottled Drinking Water in Saudi Arabia

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INTRODUCTION

Per- and polyfluoroalkyl substances (PFAS) are a broad class of synthetic chemicals that have been used extensively since the 1940s because of properties such as resistance to heat, water, and oil [1]. This class encompasses thousands of compounds, of which perfluorooctanoic acid (PFOA) and perfluorooctanesulfonic acid (PFOS) are among the most extensively studied. Due to their prevalent application in industrial settings and consumer goods—such as firefighting foams, nonstick cookware, food packaging, textiles, and electronics—PFAS have emerged as pervasive environmental pollutants [2].

A defining characteristic of many PFAS is their environmental persistence. Often referred to as "forever chemicals," they resist breakdown under natural environmental conditions [3]. This chemical resilience stems from the robust carbon-fluorine bonds, rendering them impervious to degradation from heat, light, and microorganisms. Consequently, PFAS can linger in soil, water, and air for prolonged durations, resulting in bioaccumulation and biomagnification within ecosystems [4]. The environmental spread of PFAS is not confined to polluted sites or industrial areas. Studies have revealed the existence of PFAS in diverse matrices globally, encompassing remote regions, groundwater, surface water, sediments, and even wildlife in the Arctic [5]. In East Asia, for instance, both historical and newly identified PFAS have been found in aquatic environments, indicative of local industrial practices and inadequate regulatory measures [6].

Human exposure to PFAS occurs through various pathways, including the ingestion of contaminated drinking water and food, the use of consumer products, and occupational activities [7]. A study conducted in two Asian nations revealed significant PFAS contamination in both drinking and source water, highlighting the global nature of exposure risks [8]. Once introduced into the human body, PFAS can attach to proteins and accumulate in the blood, liver, and kidneys, persisting in the system for years [9]. The health effects linked to PFAS exposure are varied and alarming. Epidemiological studies have associated PFAS exposure with a range of negative health outcomes, including immune system suppression, thyroid dysfunction, developmental delays in children, elevated cholesterol levels, and certain cancers [7]. The United States EPA has indicated that there is no known safe level of exposure for some PFAS compounds, reflecting the increasing consensus regarding their toxicity [1].

Governments and regulatory agencies globally have started to address the PFAS crisis by establishing limits, revising guidelines, and phasing out specific compounds. In Europe, regulatory advancements have concentrated on restricting PFAS usage under REACH regulations, with a growing emphasis on group-based assessments rather than evaluating PFAS compounds in isolation [10]. Concurrently, the U.S. EPA has formulated health advisories and analytical methods such as Method 537.1 for detecting selected PFAS in drinking water [11-12]. These initiatives are complemented by efforts from international organizations like the Codex Alimentarius and the World Health Organization (WHO), which are investigating the incorporation of PFAS into food safety standards. The Association of Official Analytical Collaboration (AOAC) has released Standard Method Performance Requirements (SMPRs) for PFAS in food, signifying a transition towards more standardized and dependable testing protocols [13].

The limited availability of approved analytical techniques for PFAS is a result of the challenges associated with their detection. Currently, over 13,000 distinct PFAS compounds have been recognized [1], many of which are found at ultra-trace concentrations in environmental samples. Their chemical variability and strong attraction to laboratory surfaces create additional analytical challenges, necessitating the use of highly sensitive and selective detection methods [14]. Furthermore, contamination arising from sampling containers and laboratory equipment is a well-

established issue during PFAS monitoring. These obstacles have led to the creation of robust, validated methodologies for analyzing environmental and food samples.

Liquid chromatography (LC) techniques, especially when combined with mass spectrometry (LC-MS or LC-MS/MS), are considered the benchmark for PFAS analysis due to their sensitivity, specificity, and broad applicability. EPA Method 537.1 and EPA Method 533 facilitate the quantification of 18–25 different PFAS in drinking water, while the U.S. Food and Drug Administration (FDA) has developed analogous methods for PFAS in food and feed [11,12]. LC-based methods such as high-performance liquid chromatography (HPLC), ultra-high-performance liquid chromatography (UHPLC), and capillary liquid chromatography (CLC) are particularly effective for ionic PFAS, including perfluoroalkyl carboxylic acids (PFCAs) and perfluoroalkyl sulfonic acids (PFSAs) [15]. In contrast, gas chromatography–mass spectrometry (GC-MS) is utilized for more volatile PFAS following chemical derivatization, such as fluorotelomer alcohols (FTOHs) and perfluorinated sulfonamido ethanols (FASEs) [16].

All chromatography-based techniques necessitate comprehensive sample preparation to mitigate matrix interferences and enhance accuracy. Solid-phase extraction (SPE) and liquid–liquid extraction (LLE) are frequently utilized processes, especially for aqueous matrices [15]. These validated methodologies, as detailed in ASTM D7979-17 and ASTM D7968-17a, are crucial for ensuring the quality of data [12].

Recent advancements have led to the emergence of non-traditional methods for monitoring PFAS. Sensor-based technologies and portable devices are becoming valuable supplementary tools for real-time detection of PFAS in water, providing advantages such as rapidity, cost efficiency, and suitability for field use [16]. High-resolution mass spectrometry (HRMS) has also attracted interest due to its potential for non-targeted screening and the identification of new PFAS compounds, particularly those that are intended to replace older substances [17]. Nevertheless, these innovative techniques continue to encounter obstacles related to detection limits, method verification, and practical deployment.

The present study applies an established SPE-LC-MS/MS approach based on U.S. EPA Method 537.1 to investigate the occurrence of selected PFAS in commercially available bottled drinking water in Saudi Arabia. The novelty of this study lies primarily in generating regional occurrence data for PFAS in bottled-water brands available in the Saudi market, rather than in developing a new analytical methodology. By addressing the limited availability of regional PFAS occurrence data, this study contributes preliminary evidence on the presence and concentrations of selected PFAS in the analyzed bottled-water products. These findings may support future PFAS monitoring, exposure and risk assessment, and regulatory surveillance programs in Saudi Arabia

Although PFAS are environmental contaminants rather than functional ingredients or bioactive compounds, their monitoring in drinking water remains relevant from a food-safety and quality-assurance perspective. Drinking water represents an important potential pathway of human exposure to PFAS, and their occurrence has been reported in both tap and bottled water worldwide [5]. The detection of PFAS in commercially available bottled water further highlights the importance of including drinking water in monitoring programs to support consumer safety and water-quality assessment [8]. Monitoring persistent contaminants such as PFAS contributes to contaminant surveillance, raw-material and ingredient safety, quality control, risk assessment, regulatory compliance, and consumer protection. Accordingly, the relevance of the present study to Functional Food Science lies primarily in food and beverage safety and quality assurance rather than in the evaluation of functional ingredients or physiological functionality.

MATERIAL

Sample collection: Nineteen commercially available bottled drinking-water samples, representing 19 different brands, were purchased from retail markets in Riyadh, Saudi Arabia, during the second quarter of 2025. The brands were selected to capture a range of bottled-water products available in the surveyed Riyadh retail market at the time of sample collection. One sample from each brand was obtained and transported to the laboratory for analysis. In the laboratory, the samples were inspected for physical abnormalities and homogenized prior to extraction and PFAS analysis. Given that only one sample per brand was collected from retail outlets in Riyadh, this sampling design represents a preliminary occurrence survey of PFAS in the tested bottled-water products. The findings should therefore not be interpreted as estimates of national PFAS prevalence, brand-wide contamination, or regulatory compliance across Saudi Arabia.

Chemicals and Reagents: A certified PFAS reference-standard mixture containing 24 components (100 µg/mL in methanol; Ref. RD0925561, CPAchem, Bulgaria) was used in this study. The mixture consisted of 12 native PFAS compounds and 12 corresponding isotopically labeled analogues. Of the 12 native PFAS compounds included in the reference mixture, ten were selected as target analytes for quantitative evaluation in the present study: PFHxA, NMeFOSAA, PFHpA, PFOA, PFUnA, PFDoA, PFNA, PFDA, PFTA, and PFTrDA. The corresponding isotopically labeled analogues contained in the certified PFAS standard mixture were used as internal standards. The isotopically labeled internal standards were added consistently to the calibration standards, quality-control samples, and sample extracts at the same fortification level to support analyte identification and quantitative LC-MS/MS analysis.

High-performance liquid chromatography (HPLC) grade methanol (CAS No. 67–56–1), Ammonium acetate (CAS No. 631–61–8) and acetonitrile (CAS No. 75–05–8) were acquired from Supelco, Merch, Darmstach - Germany. Ultrapure water was obtained from an Milli-Q® Direct Water Purification System. BRAND® microcentrifuge tubes, 1.5mL with lid, polypropylene Product No. BR780420 and Corning® Falcon® polypropylene Centrifuge Tubes UNSPSC Code: 41121700 No. CLS352070 were from Supelco, Merch. Sample extraction was performed using solid-phase extraction (SPE) cartridges ISOLUTE 101 500 mg/6ml Part No. 101-0050-C specifically designed for fluorinated alkyl substances was acquired from Biotage, UK.

METHODS

Analytical Method: In our study, we selected Method 537.1, an EPA-approved method for PFAS analysis that requires a solid-phase extraction (SPE) step of polystyrene divinylbenzene (SDVB) to concentrate the sample, followed by analysis using LC-MS/MS equipped with a C18 column. The limits of detection (LOD) for Method 537.1 range from 0.71 to 2.8 parts per trillion [18].

EPA Method 537.1 was utilized, which involves solid phase extraction (SPE) and liquid chromatography-tandem mass spectrometry (LC-MS/MS) for the detection of ten selected PFAS compounds in drinking water. The chromatographic separation was conducted using a C18 analytical column. The LC-MS/MS system was configured with multiple reaction monitoring (MRM) to ensure specific detection. Methanol, water, and acetonitrile were used as solvents. To reduce sample contamination, a delay column was added, and all internal tubing was replaced. Table 1 outlines the analytical conditions used for PFAS separation.

Instrumental Setup: The analysis was performed using a Shimadzu LCMS-8060 triple-quadrupole liquid chromatography-tandem mass spectrometry (LC-MS/MS) system (Shimadzu Corporation, Japan) equipped with an electrospray ionization (ESI) source operated in negative-ion mode. Data acquisition was performed using multiple reaction monitoring (MRM) for the

selective detection and quantification of the target PFAS compounds. The precursor-to-product ion transitions used for MRM detection of the target PFAS were based on those reported in Table 4 of U.S. EPA Method 537.1 (Version 2.0).

Chromatographic separation was achieved using an SP-C18 analytical column (2.1 mm × 150 mm, 2.7 µm) together with a GIST C18 delay column (3.0 mm × 50 mm, 5 µm). Mobile phase A consisted of 5 mM ammonium acetate in water, while mobile phase B consisted of methanol. The chromatographic gradient was applied over an 18-min run time at a flow rate of 0.25 mL/min. The injection volume was 2 µL, and the column oven and sample cooler temperatures were maintained at 45°C and 15°C, respectively. The detailed chromatographic conditions are presented in Table 1.

Table 1: Analytical conditions employed for the separation of PFAS compounds in drinking water samples.

Analytical Column	SP-C18, (2.1 mmI.D., x 150 mmL., 2.7 µm)
Delay Column	GIST C18 (3.0 mmI.D. x 50 mmL, 5 µm)
Mobile Phase A	5 mM ammonium acetate in water
Mobile Phase B	Methanol
Gradient Program	0 - 1.0 minutes: 95% A 5% B 1.0 - 8.0 minutes: 60% A 40% B 8.0 - 8.10 minutes: 0% A 100% B 8.10 – 13.0 minutes: 0% A 100% B 13.0 – 13.10 minutes: 95% A 5% B 13.10 – 18.0 minutes: 95% A 5% B
Flow Rate	0.25 mL/min
Injection volume	2 µL
Column Oven Temperature	45 °C

Standard Solutions Preparation: Ten calibration standards were prepared in methanol/water (96:4, v/v) at concentrations of 0.1, 0.2, 0.5, 1, 5, 10, 25, 50, 100, and 200 ng/L. All solutions were stored in polypropylene containers to minimize PFAS contamination. Materials used were pre-rinsed with methanol and distilled water before use. Table 2 summarizes the calibration standards and their preparation volumes.

Table 2: Calibration standards and corresponding preparation volumes used for PFAS analysis.

Standards	Concentration (ng/L)	V(ul) intermediate #3	V(ul) intermediate #2	V(ul) intermediate #1	v(ml) MeOH/H_2O 96/4	V(ul)
PFHxA	0.1	4			1996	2000
NMeFOSAA	0.2	8			1992	2000
PFHpA	0.5	20			1980	2000
PFOA	1	40			1960	2000
PFUnA	5		20		1980	2000
PFDoA	10		40		1960	2000
PFNA	25			20	1980	2000
PFDA	50			40	1960	2000
PFTA	100			80	1920	2000
PFTrDA	200			160	1840	2000

Sample Extraction: A 250-mL aliquot of each bottled-water sample was subjected to solid-phase extraction (SPE) following the extraction procedure described in U.S. EPA Method 537.1. SPE was performed using ISOLUTE 101 cartridges (500 mg/6 mL; Biotage, UK). Prior to sample loading, each SPE cartridge was cleaned and conditioned with 15 mL of methanol followed by 18 mL of reagent water, without allowing the sorbent bed to dry during the conditioning process. An additional 2–3 mL of reagent water was added to the cartridge before sample loading. The water sample was then passed through the conditioned cartridge under vacuum at an approximate flow rate of 10–15 mL/min, while ensuring that the cartridge did not dry before the entire sample had passed through.

Following sample loading, the sample container was rinsed twice with 7.5-mL aliquots of reagent water, and each rinse was passed through the SPE cartridge. Air or nitrogen was subsequently drawn through the cartridge for 5 min under high vacuum (10–15 in. Hg) to dry the cartridge. The retained analytes were eluted by rinsing the sample container with 4 mL of methanol and passing the methanol through the cartridge under low vacuum at a dropwise flow rate. This elution step was repeated with a second 4-mL aliquot of methanol.

The resulting extract was concentrated to dryness under a gentle stream of nitrogen in a heated water bath at 60–65°C. The dried extract was reconstituted with 96:4 (v/v) methanol:water to a final volume of 1 mL and vortex-mixed prior to LC-MS/MS analysis. An aliquot of the final extract was transferred to a polypropylene autosampler vial for instrumental analysis. Figure 1 provides a schematic representation of the extraction procedure.

Each commercial bottled-water sample was subjected to a single SPE extraction and LC-MS/MS analysis as part of the occurrence survey. Repeatability was assessed separately during method verification through the analysis of six replicate distilled-water samples fortified with the target PFAS at 10 ng/L, as described in Section 3.1.3.

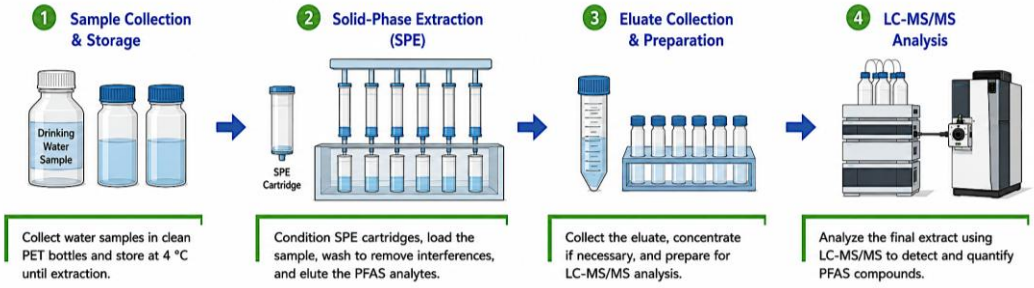


Figure 1. Schematic representation of the sample extraction procedure used for PFAS analysis.

RESULTS AND DISCUSSION

Method Verification: The established SPE-LC-MS/MS procedure based on U.S. EPA Method 537.1 was verified under the laboratory conditions used in this study prior to its application to bottled-water samples. Analytical performance was assessed in terms of selectivity and specificity, calibration range and linearity, repeatability, recovery, limits of detection (LOD) and quantification (LOQ), and quality-control samples. These experiments were intended to verify the suitability of the established analytical procedure for the determination of the selected PFAS under the conditions of the present study rather than to validate a newly developed analytical method [19–21].

Selectivity and specificity: Selectivity and specificity are critical aspects of analytical method Verification, ensuring that the method can accurately distinguish and

quantify target PFAS compounds in the presence of potential interferences. In this study, Method 537.1 was employed, which is designed to provide high selectivity for a broad range of PFAS analytes in drinking water matrices. The method's specificity was demonstrated by analyzing samples containing the targeted analytes and blank samples to assess if the method can accurately measure the PAFAS compounds.

Range of linearity: Linearity in an analytical method indicates the ability of the analytical response to vary proportionally with analyte concentration within a defined range. Calibration curves were established using standard solutions at concentrations of 0.1, 0.2, 0.5, 1, 5, 10, 25, 50, 100, and 200 ng/L. Calibration performance was evaluated in accordance with the acceptance criteria described in U.S. EPA Method 537.1. Each calibration standard was back-calculated against the calibration curve, with acceptable calculated concentrations required to be within 70–130% of the nominal concentration, except for the lowest calibration standard, for which 50–150% was considered acceptable. Linearity was evaluated using the coefficient of determination (R^2). Most target analytes showed R^2 values greater than 0.99, whereas PFUnA, PFDA, and PFNA showed R^2 values below 0.99. These results indicate variable calibration performance among the target PFAS and should be considered when interpreting the quantitative results (Figure 2).

Accuracy

a. Repeatability: Repeatability was evaluated by analyzing six replicate distilled-water samples fortified with the target PFAS compounds at 10 ng/L (Table 3, Figure 2). The analyses were performed by a single analyst under the same analytical conditions. Repeatability performance was evaluated against the acceptance criteria specified in U.S. EPA Method 537.1, with an RSD of <20% and a mean recovery within 70–130% of the fortified concentration considered acceptable. Recovery performance varied among the ten target PFAS compounds. Five analytes (PFHxA, NMeFOSAA, PFHpA, PFOA, and PFDoA) met both the recovery and RSD acceptance criteria. In contrast, PFUnA (146.0% recovery; 23.2% RSD), PFNA (132.7%; 23.3%), and PFDA (143.9%; 25.4%) did not meet either the recovery or RSD acceptance criteria. PFTA (51.0%; 5.5%) and PFTrDA (65.2%; 6.3%) met the RSD criterion but did not meet the recovery criterion. These deviations indicate analyte-dependent differences in analytical performance and should be considered when interpreting the quantitative concentrations reported for the affected compounds.

Table 3: Repeatability results for distilled-water samples fortified with PFAS at 10 ng/L (n = 6)

Compound	Average concentration (ng/L)	Recovery (%)	Standard deviation (SD)	RSD (%)
PFHxA	9.84	98.4	1.69	17.2
NMeFOSAA	9.41	94.1	0.86	9.1
PFHpA	10.54	105.4	1.57	14.9
PFOA	11.70	117.0	2.11	18.0
PFUnA	14.60	146.0	3.39	23.2
PFDoA	9.44	94.4	1.08	11.4
PFNA	13.27	132.7	3.09	23.3
PFDA	14.39	143.9	3.65	25.4
PFTA	5.10	51.0	0.28	5.5
PFTrDA	6.52	65.2	0.41	6.3

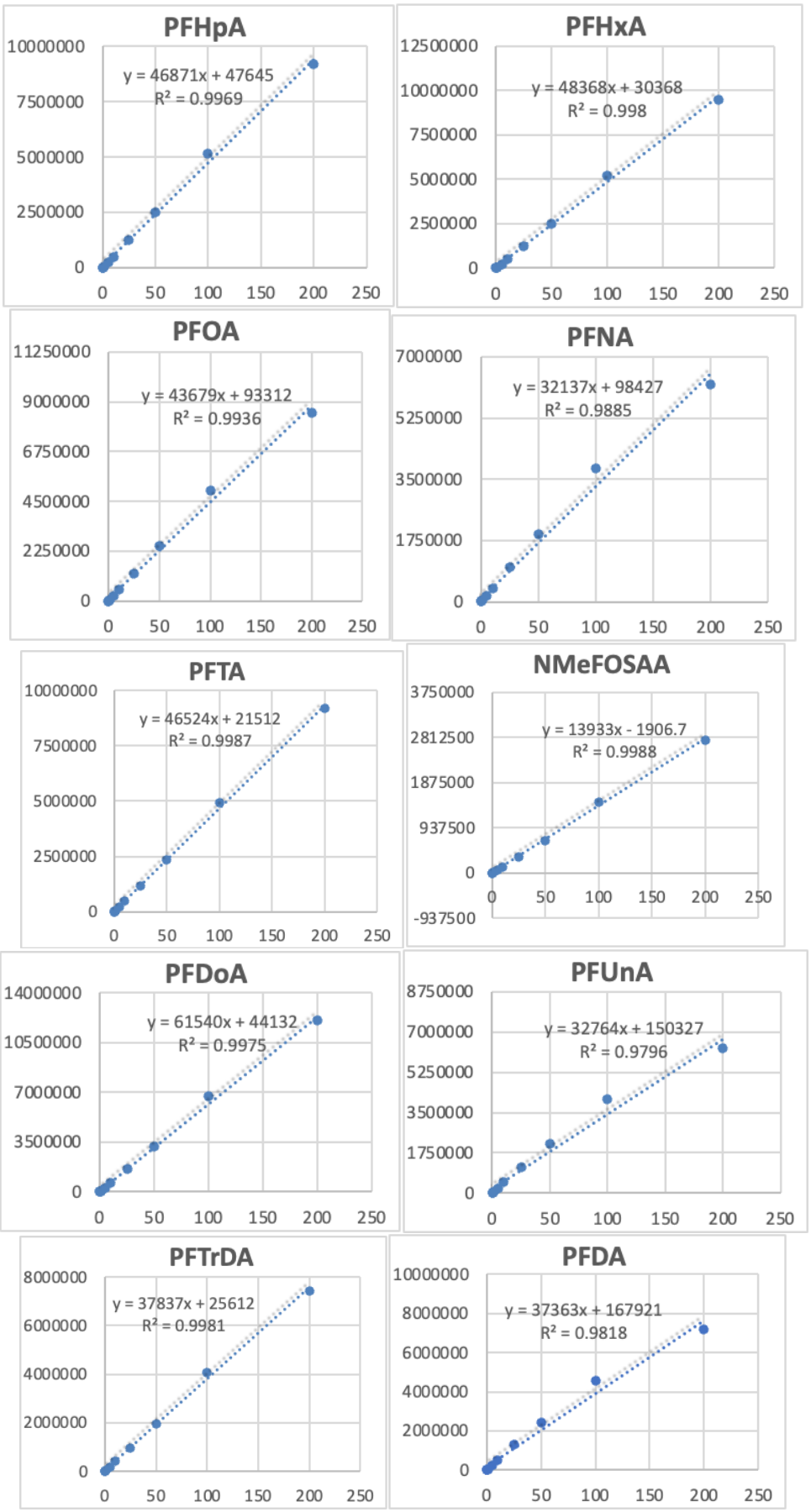


Figure 2: Standard concentrations used for PFAS compounds.

b. LOD, LOQ and Uncertainty: The limits of detection (LOD) and quantification (LOQ) were evaluated using the signal-to-noise (S/N) approach at a concentration of 0.2 ng/L. The LOD was defined as the concentration producing an S/N ratio of approximately 3, whereas the LOQ was defined as the concentration producing an S/N ratio of approximately 10. The 0.2 ng/L samples were subjected to the complete sample-preparation and SPE extraction procedure prior to LC–MS/MS analysis; therefore, the reported values represent method-level rather than instrument-only detection and quantification limits. The S/N ratio was calculated from the chromatographic response of each target analyte relative to the baseline noise in the corresponding chromatographic region. The analyte-specific LOD and LOQ results are presented in Table 4.

Measurement uncertainty was evaluated separately for each target analyte by combining the relevant uncertainty contributions associated with the analytical procedure. The combined standard uncertainty (u_c) was calculated as the square root of the sum of the squared individual standard uncertainty components:

$$u_c = \sqrt{(u_1^2 + u_2^2 + \ldots + u_n^2)}$$

The expanded uncertainty (U) was then calculated as:

$$U = k \times u_c$$

using a coverage factor of k = 2, corresponding to an approximate 95% confidence level. The analyte-specific expanded uncertainty values are reported in Table 4. [19–21].

Table 4: Limits of Detection (LOD) and Limits of Quantification (LOQ) for PFAS Compounds.

Entry	Compound	LOD (ng/L)	LOQ (ng/L)
1	PFHxA	0.04	0.11
2	NMeFOSAA	0.04	0.13
3	PFHpA	0.05	0.16
4	PFOA	0.05	0.16
5	PFUnA	0.08	0.23
6	PFDoA	0.04	0.14
7	PFNA	0.05	0.14
8	PFDA	0.09	0.27
9	PFTA	0.04	0.12
10	PFTrDA	0.07	0.21

Quality control: As a separate quality-control check, one distilled-water sample was fortified with the target PFAS compounds at 10 ng/L, while a second unfortified distilled-water sample was analyzed as a control. Both samples underwent the complete analytical procedure, including SPE extraction, drying, reconstitution, and LC-MS/MS analysis. This quality-control check was distinct from the repeatability experiment presented in Table 3 and was not part of the formal EPA Method 537.1 QC sequence.

As shown in Table 5, recoveries varied among the target analytes. PFHxA, PFHpA, PFOA, and PFNA met the 70–130% recovery criterion, whereas PFDA, NMeFOSAA, PFUnA, PFDoA, PFTrDA, and PFTA showed recoveries below 70%. These results were retained and reported transparently as part of this preliminary analytical assessment; however, the low recoveries were

considered a limitation of the method for the affected analytes, and quantitative results for these compounds were interpreted with caution. No target PFAS compounds were detected in the unfortified distilled-water control. The results are presented in Table 5.

Table 5: Quality-control results for a fortified water sample (10 ng/L) and an unfortified water sample.

PFAS Compound	Spiked sample concentration (ng/L)	Recovery (%)	Unfortified concentration (ng/L)
PFHxA	8.490	84.9	Not Detected
PFHpA	8.550	85.5	Not Detected
PFOA	9.265	92.7	Not Detected
PFNA	8.610	86.1	Not Detected
PFDA	5.823	58.2	Not Detected
NMeFOSAA	2.367	23.7	Not Detected
PFUnA	3.817	38.2	Not Detected
PFDoA	2.604	26.0	Not Detected
PFTrDA	3.404	34.0	Not Detected
PFTA	3.409	34.1	Not Detected

Using the predefined recovery acceptance range of 70–130%, PFHxA, PFHpA, PFOA, and PFNA met the recovery criterion in the fortified quality-control sample. In contrast, PFDA, NMeFOSAA, PFUnA, PFDoA, PFTrDA, and PFTA showed recoveries below the predefined acceptance range and therefore did not meet the recovery criterion. These results indicate analyte-dependent limitations in the quantitative performance of the method and should be considered when interpreting concentrations reported for the affected compounds. The absence of detectable target PFAS in the unfortified sample indicated no detectable background contribution in this quality-control check.

Analysis of Bottled Water Samples from the Saudi Market: Following application and verification of the established SPE-LC-MS/MS procedure based on U.S. EPA Method 537.1, the bottled-water samples obtained from the Riyadh retail market were analyzed for the selected PFAS compounds. In addition to PFAS analysis, the pH of each sample was measured, with values ranging from 5.6 to 6.7. The pH was recorded as a general water-quality parameter because it may influence the physicochemical behavior of PFAS in aqueous matrices and can be considered during interpretation of analytical results. However, within the observed pH range, no clear relationship was identified between pH and PFAS concentrations in the analyzed samples.

PFAS compounds were detected in the majority of the analyzed samples, generally at low and variable concentrations. For contextual comparison, U.S. EPA drinking-water MCLs were considered only for the measured PFAS compounds for which individual numerical standards are available [18]. Among the ten target analytes investigated in this study, individual U.S. EPA MCLs are available for PFOA and PFNA (Table 6) [18]. These values were used solely as regulatory benchmarks and should not be interpreted as demonstrating overall regulatory compliance of the bottled-water samples, particularly because no individual U.S. EPA MCLs are available for several of the PFAS compounds measured in this study.

One bottled-water sample (Sample No. 7; Appendix A) showed a PFOA concentration of 59.390 ng/L, which was substantially higher than the U.S. EPA drinking-water MCL of 4.0 ng/L used as a regulatory benchmark in this study [18]. However, this elevated result was obtained from a single extraction and LC-MS/MS analysis and was

not independently confirmed by repeat extraction, replicate analysis, or confirmatory injection. Therefore, the finding should be interpreted cautiously and is reported as an observed analytical result rather than a confirmed regulatory exceedance. Additional confirmatory analysis would be required to verify this elevated concentration.

In total, eight different PFAS compounds were detected across the analyzed samples. However, regulatory benchmarks were available for only a limited subset of the PFAS considered in this study, restricting comprehensive regulatory interpretation of the detected compounds. Furthermore, the analyte-dependent recovery limitations identified during method verification should be considered when interpreting quantitative concentrations for the affected PFAS compounds.

The complete PFAS concentration data for all bottled-water samples are provided in Appendix A, while Table 6 summarizes the regulatory benchmarks used for contextual comparison [18]. Overall, the findings provide preliminary occurrence data indicating the presence of PFAS at generally low and variable concentrations in commercially available bottled water sampled from the Riyadh retail market. The results support the need for broader surveillance and further studies incorporating larger and more representative sampling designs, repeated sampling, and confirmatory analysis of unusually elevated findings.

Comparison with previous PFAS occurrence studies provides useful context for the concentrations observed in the present study. In a study of 101 bottled-water products marketed in the United States, PFAS were detected in 39 products, with total PFAS concentrations ranging from 0.17 to 18.87 ng/L and 97% of the samples containing less than 5 ng/L total PFAS [22]. A recent study from southern Catalonia, Spain, reported total PFAS concentrations of up to 12 ng/L in bottled-water samples [23]. PFAS have also been investigated in tap and bottled water in Turkey, where concentrations in bottled-water samples were generally lower than those measured in tap water [24]. In addition, PFAS occurrence in drinking and source waters from the Philippines and Thailand has demonstrated that contamination levels can vary substantially among geographic regions and water sources [8]. Against this international context, the generally low concentrations observed in most samples in the present study are broadly consistent with the low-level occurrence reported in several bottled-water investigations [8,22–24], whereas the PFOA concentration of 59.390 ng/L observed in one sample was notably elevated compared with concentrations generally reported in these bottled-water studies. As noted above, however, this elevated result was not independently confirmed and should therefore be interpreted cautiously.

Direct comparison among studies should be made cautiously because reported PFAS concentrations can be influenced by differences in the number and identity of target analytes, sample preparation and extraction procedures, instrumental methods, detection and quantification limits, water sources, treatment processes, and sampling designs [5,9,14,24]. Published occurrence data specifically addressing PFAS in bottled drinking water from Saudi Arabia and the Gulf region remain limited, restricting direct regional comparison and highlighting the value of the present study as preliminary occurrence data. The present study also has important limitations. Only 19 commercially available bottled-water brands purchased from retail outlets in Riyadh were investigated, with one sample representing each brand. Consequently, the findings are limited to the analyzed sample set and should not be generalized to all bottled-water products, production batches, geographic regions, or brands marketed throughout Saudi Arabia.

Furthermore, the absence of replicate sampling across different batches or purchase locations limits assessment of within-brand variability, while the analyte-dependent recovery limitations identified during method verification introduce additional uncertainty for some quantitative results. Future studies should therefore include larger and geographically broader sampling, multiple samples and production batches per brand, and confirmatory analysis of unusually elevated findings.

Table 6: U.S. EPA drinking-water regulatory benchmarks for PFAS compounds measured in the present study for which individual maximum contaminant levels (MCLs) are available.

PFAS compound	Maximum Contaminant Level (MCL), ng/L
PFOA	**4.0**
PFNA	**10**

Note: Only PFAS compounds that were analytically measured in the present study and have an individual U.S. EPA drinking-water MCL are included in this table. PFHxS is not included because it was not among the ten target analytes measured in this study. The remaining measured PFAS compounds (PFHxA, NMeFOSAA, PFHpA, PFUnA, PFDoA, PFDA, PFTA, and PFTrDA) do not have individual U.S. EPA MCLs and therefore are not assigned numerical regulatory benchmarks in this comparison.

CONCLUSION

This study applied and verified an established SPE-LC-MS/MS procedure based on U.S. EPA Method 537.1 for the determination of selected PFAS in bottled drinking water. Analytical performance was assessed through calibration linearity, repeatability, recovery, limits of detection and quantification, and quality-control samples. The verified procedure was subsequently applied to commercially available bottled-water samples to generate regional PFAS occurrence data.

PFAS were detected in the majority of the analyzed bottled-water samples at low and variable concentrations, with eight PFAS compounds detected across the sample set. One sample contained PFOA at 59.390 ng/L, exceeding the U.S. EPA maximum contaminant level used as the regulatory comparison benchmark in this study. Because the samples were obtained from retail markets in Riyadh, these findings should be regarded as preliminary regional occurrence data and should not be interpreted as representative of all bottled drinking water marketed throughout Saudi Arabia.

The principal contribution of this study is the generation of regional occurrence data for PFAS in commercially available bottled drinking water rather than the development of a new analytical methodology. The findings support the need for continued and geographically broader PFAS surveillance and further exposure and risk-assessment studies. From a Functional Food Science perspective, monitoring persistent contaminants in drinking water also contributes to consumer safety and quality assurance, particularly because water is consumed directly and is widely used as an ingredient and processing medium in foods and beverages.

Scientific Innovations: The principal scientific contribution of this study is the generation of regional occurrence data on selected PFAS in commercially available bottled drinking water obtained from the Riyadh retail market in Saudi Arabia. Rather than developing a new analytical methodology, the study applies an established SPE-LC-MS/MS approach based on U.S. EPA Method 537.1 to address the limited availability of regional PFAS occurrence data. The detection of eight PFAS compounds across the analyzed sample set, together with the observed variability in their concentrations, provides preliminary evidence on PFAS occurrence in bottled-water products available in the

surveyed market. These findings establish a useful regional baseline for future studies involving broader geographic sampling, repeated sampling of individual brands, and PFAS exposure and risk assessment.

Practical Implications: The detection of PFAS in the majority of the 19 bottled-water samples analyzed highlights the practical importance of continued surveillance of persistent contaminants in drinking water. The findings may help inform the design of future monitoring programs by identifying PFAS occurrence in commercially available bottled water as an area requiring broader and more representative investigation. The elevated PFOA concentration observed in one sample further supports the value of follow-up monitoring and confirmatory assessment where appropriate. From a Functional Food Science perspective, PFAS surveillance in drinking water is relevant to consumer safety and quality assurance because water is consumed directly and is also widely used as an ingredient and processing medium in foods and beverages. Therefore, continued contaminant monitoring can contribute to evidence-based water-quality assessment, exposure evaluation, and consumer protection.

Abbreviations: AOAC: Association of Official Analytical Collaboration; ASTM: ASTM International; CLC: Capillary Liquid Chromatography; EPA: United States Environmental Protection Agency; ESI: Electrospray Ionization; FASEs: Perfluorinated Sulfonamido Ethanols; FDA: United States Food and Drug Administration; FTOHs: Fluorotelomer Alcohols; GC-MS: Gas Chromatography–Mass Spectrometry; HPLC: High-Performance Liquid Chromatography; HRMS: High-Resolution Mass Spectrometry; LC: Liquid Chromatography; LC-MS: Liquid Chromatography–Mass Spectrometry; LC-MS/MS: Liquid Chromatography–Tandem Mass Spectrometry; LLE: Liquid–Liquid Extraction; LOD: Limit of Detection; LOQ: Limit of Quantification; MCL: Maximum Contaminant Level; MRM: Multiple Reaction Monitoring; NMeFOSAA: N-Methyl Perfluorooctane Sulfonamidoacetic Acid; PFAS: Per- and Polyfluoroalkyl Substances; PFCA(s): Perfluoroalkyl Carboxylic Acid(s); PFDA: Perfluorodecanoic Acid; PFDoA: Perfluorododecanoic Acid; PFHpA: Perfluoroheptanoic Acid; PFHxA: Perfluorohexanoic Acid; PFNA: Perfluorononanoic Acid; PFOA: Perfluorooctanoic Acid; PFOS: Perfluorooctanesulfonic Acid; PFSA(s): Perfluoroalkyl Sulfonic Acid(s); PFTA: Perfluorotetradecanoic Acid; PFTrDA: Perfluorotridecanoic Acid; PFUnA: Perfluoroundecanoic Acid; QC: Quality Control; REACH: Registration, Evaluation, Authorisation and Restriction of Chemicals; RSD: Relative Standard Deviation; SDVB: Polystyrene-Divinylbenzene; SFDA: Saudi Food and Drug Authority; SMPRs: Standard Method Performance Requirements; SPE: Solid-Phase Extraction; UHPLC: Ultra-High-Performance Liquid Chromatography; WHO: World Health Organization.

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